

# Spanning Tree–Induced Inequalities as Chvátal–Gomory Rank-1 Cuts

Giovanni Dettori

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## 1 Motivation and Setup

Optimization problems involving polynomials with binary variables arise in many areas of discrete optimization. A general binary polynomial optimization problem can be written as

$$\min_{z \in \{0,1\}^n} f(z) = \sum_{S \subseteq [n]} c_S \prod_{i \in S} z_i, \quad (1)$$

where the coefficients  $c_S \in \mathbb{R}$  define the polynomial for every subset  $S$  of the  $n$  variables of the polynomial. Already in the quadratic case, where the objective contains only pairwise interactions, optimizing over binary variables is NP-hard [2]; allowing terms of higher degree only increases the combinatorial complexity of the problem.

These problems are usually tackled via a linearization by introducing an auxiliary variable for each multilinear term and enforcing consistency through a suitable system of linear constraints [1]. More precisely, using a hypergraph representation of monomials, let  $G = (V, E)$  be a hypergraph. Consider the multilinear polytope, defined as the convex hull of all binary points  $z \in \mathbb{Z}^{V \cup E}$  such that for every  $e \in E$ ,

$$z_e = \prod_{v \in e} z_v.$$

The standard relaxation of the multilinear polytope is given by the following linear constraints:

$$\begin{aligned} \alpha_v : \quad & z_v \leq 1 & \forall v \in V, \\ \nu_e : \quad & -z_e \leq 0 & \forall e \in E, \\ \varepsilon_e : \quad & \sum_{u \in e} z_u - z_e \leq |e| - 1 & \forall e \in E, \\ \delta_{v,e} : \quad & -z_v + z_e \leq 0 & \forall v \in e, e \in E. \end{aligned}$$

[SD] no need to put the parenthesis for the quantifiers

Relaxing the integrality condition  $z \in \mathbb{Z}^{V \cup E}$  so that  $z$  is allowed to take fractional values in the above formulation yields the LP relaxation.

The polyhedral structure of the linearized formulation has been extensively studied: in the quadratic case, this yields the Boolean quadric polytope introduced and analyzed in [4]. [SD] This sentence is not entirely correct, since you have polyhedra **only** when dealing with linear functions. So, you need to linearize the quadratic function first.

Nonetheless, the LP relaxation highlighted above can be weak, exhibiting a large integrality gap between its optimal value and the optimal value of (1). Standard algorithms for these problems operate by solving successive LP relaxations of the integer formulation and their effectiveness depends critically on how closely the initial LP relaxation approximates the convex hull of integer-feasible solutions. Providing algorithms with a tight relaxation to start with is therefore fundamental.

[SD] you haven't mentioned yet what is the goal, so "the following analysis" is a bit vague.

## 2 Main Results

### 2.1 Notations and Definitions

**Definition 1** (Intersection graph). *Let  $G = (V, E)$  be a hypergraph, and let  $F \subseteq E$  be a nonempty subset of its hyperedges. The intersection graph induced by  $F$  in  $G$ , denoted  $I_G(F)$ , is the undirected graph*

$$I_G(F) = (V_I, E_I),$$

where

$$V_I := F,$$

$$E_I := \{ \{f', e'\} \subseteq F \mid f' \neq e', f' \cap e' \neq \emptyset \}.$$

*Thus, each node of  $I_G(F)$  corresponds to a hyperedge of  $G$ , and two nodes  $f', e' \in F$  are adjacent in  $I_G(F)$  if and only if the corresponding hyperedges intersect in  $G$ . For every  $\{f', e'\} \in E_I$ , we fix a representative intersection vertex  $v_{f'e'} \in f' \cap e'$  to label the edge  $(f', e')$  of  $I_G(F)$ .*

**Definition 2** (Partial hypergraph). *Let  $G = (V, E)$  be a hypergraph and let  $E' \subseteq E$  be a subset of its hyperedges. The partial hypergraph induced by  $E'$  is*

$$H := (V', E'), \quad V' := \bigcup_{e \in E'} e.$$

*In other words,  $H$  contains exactly the hyperedges in  $E'$  together with all vertices incident to them.*

**Definition 3** (Berge-cycle,  $\gamma$ -cycle,  $\beta$ -cycle). Let  $G = (V, E)$  be a hypergraph. A Berge-cycle of length  $m \geq 2$  is a sequence

$$v_1, e_1, v_2, e_2, \dots, v_m, e_m, v_1$$

such that

- $v_1, v_2, \dots, v_m$  are distinct vertices,
- $e_1, e_2, \dots, e_m$  are distinct hyperedges, and
- $v_i, v_{i+1} \in e_i$  for all  $i = 1, \dots, m-1$ , and  $v_m, v_1 \in e_m$ .

A hypergraph is Berge-acyclic if it contains no Berge-cycles.

A Berge-cycle is called a  $\gamma$ -cycle if  $m \geq 3$  and each vertex  $v_i$  with  $i = 2, \dots, m$  belongs exactly to the two consecutive cycle edges  $e_{i-1}$  and  $e_i$ , and to no other edge of the hypergraph.

A  $\beta$ -cycle is a  $\gamma$ -cycle with the additional requirement that also the vertex  $v_1$  belongs only to the two edges  $e_1$  and  $e_m$ , and to no other edge.

## 2.2 Spanning-Tree Inequality

**Remark 1** (Assumptions). Let  $G = (V, E)$  be a hypergraph. Take  $f \in E$  with  $|f| \geq 2$  and let  $F \subseteq E \setminus \{f\}$  be a nonempty subset of hyperedges. Consider the partial hypergraph

$$H = (V(H), F \cup \{f\}), \quad \text{where } V(H) := \bigcup_{e \in F \cup \{f\}} e.$$

We assume that  $H$  contains a  $\gamma$ -cycle

$$C = v_1, e_1, v_2, e_2, \dots, v_m, e_m, v_1, \quad \text{with } e_1 = f.$$

Let  $T$  be any spanning tree of the intersection graph  $I_H(F)$  induced by  $F$ .

Finally, we define the uncovered and covered vertices of  $f$  with respect to  $F$  as

$$P_f := f \setminus \bigcup_{e \in F} e, \quad C_f := f \setminus P_f.$$

For each  $v \in C_f$ , choose and fix one arbitrary hyperedge  $e_v \in F$  such that  $v \in e_v$ . Such a choice is always possible by the definition of  $C_f$ . We denote by

$$E_{C_f} := \{e_v : v \in C_f\}$$

the collection of chosen hyperedges (note that the same hyperedge may appear multiple times).

**Theorem 1** (Tree-spanning inequality). *With the notation and assumptions above, the inequality*

$$\boxed{\sum_{e \in F} z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} - z_f + \sum_{v \in P_f} z_v \leq |P_f|} \quad (2)$$

*is valid for the multilinear polytope and it is a Chvátal–Gomory cut of rank 1.*

*Proof.* Set  $\lambda := 1/|C_f|$ . Take  $(1 - \lambda) \cdot \nu_f$  and  $\lambda \cdot \varepsilon_f$ :

$$(1 - \lambda)(-z_f) + \lambda \left( \sum_{u \in f} z_u - z_f \right) \leq \lambda(|f| - 1).$$

resulting in

$$-z_f + \lambda \sum_{u \in f} z_u \leq \lambda(|f| - 1). \quad (3)$$

For each  $v \in C_f$ , add  $\lambda \cdot \delta_{v, e(v)} : -\lambda z_v + \lambda z_{e(v)} \leq 0$  where  $e_v \in E_{C_f}$  denotes the (fixed) hyperedge of  $F$  containing  $v$  (as defined in Remark 1,  $E_{C_f}$  is the set of chosen hyperedges associated with the vertices in  $C_f$ ).

Summing over  $C_f$  yields

$$-z_f + \lambda \sum_{w \in P_f} z_w + \lambda \sum_{v \in C_f} z_{e(v)} \leq \lambda(|f| - 1). \quad (4)$$

For each  $w \in P_f$  add  $(1 - \lambda) \cdot \alpha_w : (1 - \lambda)z_w \leq (1 - \lambda)$ . Summing over  $P_f$  raises each uncovered vertex from coefficient  $\lambda$  to 1 and contributes  $(1 - \lambda)|P_f|$  to the right-hand side:

$$-z_f + \sum_{w \in P_f} z_w + \lambda \sum_{v \in C_f} z_{e(v)} \leq (1 - \lambda)|P_f| + \lambda(|f| - 1). \quad (5)$$

On a separate note, for each edge  $(f', e') \in T$  we add the inequality

$$\delta_{v_{f' e'}, e'} : -z_{v_{f' e'}} + z_{e'} \leq 0,$$

and sum over all edges of  $T$ . Since each hyperedge  $e \in F$  appears in the tree with degree  $\deg_T(e)$  (i.e., the number of edges of  $T$  incident to  $e$ ), this yields

$$\sum_{(f', e') \in T} (-z_{v_{f' e'}} + z_{e'}) = \sum_{e \in F} \deg_T(e) z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} \leq 0.$$

For each  $e \in F$  we now add  $(\deg_T(e) - 1) \cdot \nu_e : (\deg_T(e) - 1)(-z_e) \leq 0$ . [\[SD\] summing with integer multipliers leads to dominated inequalities...](#) Summing over all  $e \in F$  normalizes the coefficients of the  $z_e$ -variables and gives

$$\sum_{e \in F} z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} \leq 0.$$

Finally, for each  $v \in C_f$  we add  $\lambda \cdot \nu_{e(v)} : -\lambda z_{e(v)} \leq 0$ , and summing over all  $v \in C_f$  yields

$$\sum_{e \in F} z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} - \lambda \sum_{v \in C_f} z_{e(v)} \leq 0. \quad (6)$$

Adding (5) to (6) yields

$$\sum_{e \in F} z_e - \sum_{(f', e') \in T} z_{v_{f' e'}} - z_f + \sum_{w \in P_f} z_w \leq (1 - \lambda)|P_f| + \lambda(|f| - 1).$$

Since  $f = P_f \cup C$  where the union is disjoint, we have

$$|f| = |P_f| + |C_f|.$$

As  $\lambda = 1/|C_f|$ , the right-hand side equals

$$(1 - \lambda)|P_f| + \lambda(|P_f| + |C_f| - 1) = |P_f| + \lambda(|C_f| - 1) = |P_f| + \frac{|C_f| - 1}{|C_f|}.$$

Given the assumptions,  $|C_f| \geq 2$  and therefore:

$$0 < \frac{|C_f| - 1}{|C_f|} < 1.$$

Therefore, taking the floor of the RHS gives exactly (2). All multipliers used before rounding are non-negative and since we only use inequalities of the standard linearization, (2) is a rank-1 CG cut.  $\square$

[SD]

**To do:**

- create a folder with examples of hypergraph and associated facets/intersection inequalities (it would be nice to have a big enough set in order to test different scenarios)
- concerning the redundancy issue: generate for a few examples **all** the intersection inequalities and check how many of them are facet-defining (at least one meaningful example)

## 2.3 Connections to Odd- $\beta$ Cycle Inequalities

A subset of Odd- $\beta$  cycle inequalities as part of the Spanning Tree inequality Odd  $\beta$ -cycle inequalities are defined in [5]. Some of them are contained within the family of spanning tree inequalities. In particular,

if  $G'$  contains odd  $\beta$ -cycles and the spanning-tree generating (2) is a path, then the inequality is an odd  $\beta$ -cycle inequality.

*Proof.* The beta-cycle inequalities are defined in [5] as follows:

Consider a hypergraph  $G = (V, E)$ , let  $C = v_1, e_1, v_2, \dots, v_m, e_m, v_1$  be a  $\beta$ -cycle in  $G$ , and let  $E^-, E^+$  be a partition of  $E(C)$  such that  $k := |E^-|$  is odd and  $e_1 \in E^-$ . Let  $D := \{e_{p+1}, e_{p+2}, \dots, e_m\}$ , where  $e_p$  is the last edge in  $C$  that belongs to  $E^-$ . We denote by  $f_1, \dots, f_k$  the subsequence of  $e_1, \dots, e_m$  consisting of the edges in  $E^-$ . Let  $S_1 := (\cup_{e \in E^-} e) \setminus (\cup_{e \in E^+} e)$  and  $S_2 := V(C) \setminus (\cup_{e \in E^-} e)$ . With this notation in place, we make the following assumptions:

1. Every node  $v \in \cup_{i=1}^m e_i$  is contained in at most two edges among  $e_1, \dots, e_m$ .
2. For every edge  $e_i \in E^+ \setminus D$ , every edge in  $E^-$  adjacent to  $e_i$  (if any) is either  $e_{i-1}$  or  $e_{i+1}$ .
3. No edge in  $D$  is adjacent to an edge  $f_i$  with  $i$  even.
4. At least one of the following two conditions holds:
  - (a) For every  $v \in S_1$ , either  $v$  is contained in just one edge  $e \in E^-$ , or it is contained in two edges  $f_i, f_j$  with  $i$  odd and  $j$  even.
  - (b) For every  $e' \in E^-$  and  $e'' \in D$  such that  $e' \cap e'' \neq \emptyset$ , then either  $e' = e_1$ ,  $e'' = e_m$  or  $e' = e_p$ ,  $e'' = e_{p+1}$ .

We then define the odd  $\beta$ -cycle inequality corresponding to  $C$  and  $E^-$  as

$$\sum_{v \in S_1} z_v - \sum_{e \in E^-} z_e - \sum_{v \in S_2} z_v + \sum_{e \in E^+} z_e \leq |S_1| - |\{i \in \{1, \dots, m\} : e_i, e_{i+1} \in E^-\}| + \left\lfloor \frac{k}{2} \right\rfloor. \quad (7)$$

Before proceeding in the proof, it is necessary to check some notation: in the spanning tree construction,  $E^-$  coincides with  $f$  hence its cardinality  $k$  is always odd and equal to 1.  $E(C) \setminus f$  is necessarily  $E^+$  and  $D$  collapses to  $E^+$ ; also,  $S_1$  equals  $P_f$ . Finally, every hyperedge  $e \in E^+$  appears as a node of the intersection graph, and two hyperedges  $p, e$  are connected if they share some vertex  $v_{pe}$ . If  $T$  is a path, the hyperedges  $(f', e') \in E^+$  (that are now vertices in  $T$ ) are adjacent in the  $\beta$ -cycle, and the intersection vertex  $v_{f'e'} \in f' \cap e'$  is a vertex connecting them in the original hypergraph. Hence, the set

$$\{v_{f'e'} : (f', e') \in T\}$$

is precisely the set of vertices that: belong to the  $\beta$ -cycle (since they lie in two consecutive hyperedges of it) and do not belong to  $f$ . But those two conditions define  $S_2$  exactly.

We now verify that assumptions (1)–(4a) from the definition of Odd- $\beta$ -cycle inequalities are satisfied by the spanning tree construction introduced in Theorem 1.

**Assumption (1).** The  $\beta$ -cycle corresponding to the spanning tree inequality is generated by taking all hyperedges in  $E(C) = E^- \cup E^+$ , with  $E^- = \{f\}$  and  $E^+ = F$ . In this case, the spanning tree  $T$  is a path. Therefore, every vertex appearing in the  $\beta$ -cycle is incident to at most two edges in the path, so assumption (1) is satisfied.

**Assumption (2).** By construction,  $D = \{e_{p+1}, \dots, e_m\}$  collects the hyperedges of the cycle that appear after the last hyperedge in  $E^-$ . Since in our setting  $E^- = \{f\}$  and  $f = e_1$ , we have  $p = 1$  and consequently  $D = E^+ = E \setminus \{f\}$  the condition in assumption (2) becomes irrelevant, as  $E^+ \setminus D$  is empty, and the adjacency requirement is trivially satisfied.

**Assumption (3).** Assumption (3) states that no hyperedge in  $D$  is adjacent to an edge  $f_i$  with even index. In our construction,  $E^- = \{f_1\}$  consists of a single hyperedge, so  $i = 1$  is odd and no even-indexed negative hyperedge exists. Therefore, assumption (3) is automatically fulfilled.

**Assumption (4a).** For assumption (4a) we have  $S_1 = P_f$  and vertex  $v \in P_f$  is, by definition, a vertex of  $f$  not contained in any other hyperedge of the cycle. Hence, every  $v \in S_1$  belongs to exactly one hyperedge  $e \in E^-$ . This satisfies the first clause of assumption (4a), and thus assumption (4a) holds.

Consequently, the spanning tree (path) construction satisfies assumptions (1)–(4a) of the odd- $\beta$ -cycle inequality definition. In this special case, the LHS resulting from the odd- $\beta$ -cycle coincides with the spanning tree inequality (2), as shown by the identification  $S_1 = P_f$ ,  $S_2 = \{v_{f'e'} : (f', e') \in T\}$ ,  $E^+ = F$  and  $E^- = f$ . The RHS of (7) shows 3 terms where:

1.  $|S_1| = |P_f|$ .
2.  $|\{i \in \{1, \dots, m\} : e_i, e_{i+1} \in E^-\}| = 0$ .
3.  $\lfloor \frac{k}{2} \rfloor = \lfloor \frac{1}{2} \rfloor = 0$ .

Thus the odd  $\beta$ -cycle inequality (7) coincides exactly with the spanning tree inequality (2) in this special case.

□

### 3 Separation problem for Spanning–Tree Inequalities

Given a fractional point  $\bar{z} \in \mathbb{R}^{V \cup E}$  obtained from the LP relaxation described in the setup, the separation problem for the spanning–tree inequalities introduced in Theorem 1 is the following:

*Determine whether there exist  $f \in E$ , a nonempty family  $F \subseteq E \setminus \{f\}$  of hyperedges, and a spanning tree  $T$  of the intersection graph  $I_H(F)$  such that the corresponding spanning–tree inequality is violated by  $\bar{z}$ .*

Equivalently, we seek  $f, F, T$  for which the violation

$$\psi(f, F, T; \bar{z}) := \sum_{e \in F} \bar{z}_e - \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}} - \bar{z}_f + \sum_{v \in P_f} \bar{z}_v - |P_f| \quad (8)$$

is strictly positive. Here  $P_f := f \setminus \bigcup_{e \in F} e$  denotes the uncovered vertices of  $f$ . A positive value of  $\psi(f, F, T; \bar{z})$  means that the spanning-tree inequality (2) cuts off the fractional point  $\bar{z}$ .

The separation problem can be decomposed into two layers:

- (i) an *outer* combinatorial search over the choices of  $f$  and  $F$ , and
- (ii) an *inner* optimization over all spanning trees  $T$  of the intersection graph  $I_H(F)$ .

Only the second layer depends on the choice of the tree  $T$ . In order to isolate this dependence, we rewrite the violation in a form separating the contributions of  $f$  and  $F$  from that of  $T$ .

The last two terms of (8) can be grouped as

$$\sum_{v \in P_f} \bar{z}_v - |P_f| = \sum_{v \in P_f} (\bar{z}_v - 1),$$

so that

$$\psi(f, F, T; \bar{z}) = \sum_{e \in F} \bar{z}_e - \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}} - \bar{z}_f + \sum_{v \in P_f} (\bar{z}_v - 1). \quad (9)$$

This shows that each uncovered vertex  $v \in P_f$  contributes the term  $(\bar{z}_v - 1)$  to the violation. In particular, all contributions of  $f$  and of the uncovered vertices  $P_f$  are independent of the choice of the spanning tree  $T$ .

For fixed  $f$  and  $F$ , the only term in (9) that depends on  $T$  is

$$- \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}}.$$

### 3.1 Reduction to a minimum spanning tree

Given the above, we argue that maximizing  $\psi(f, F, T; \bar{z})$  over all spanning trees  $T$  is equivalent to minimizing

$$S(T) := \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}}.$$

This is exactly the problem of computing a minimum-weight spanning tree (MST) of the intersection graph  $I_H(F)$ , where each edge  $(f', e')$  has weight  $w(f', e') := \bar{z}_{v_{f'e'}}$ .

By the classical analysis of Kruskal's and Prim's algorithms (see [3, Chapter 4.5]), an MST can be computed in  $O(|E_I| \log |F|)$  time. Since any two hyperedges in  $F$  may intersect, the intersection graph

$I_H(F)$  can contain up to  $\binom{|F|}{2}$  edges, that is, a number of edges quadratic in  $|F|$ . Consequently, computing an MST requires  $O(|F|^2 \log |F|)$  time in the worst case. Let

$$S(\text{MST}(I_H(F))) = S(T^*) := \min_{T \text{ spanning tree of } I_H(F)} \sum_{(f', e') \in T} \bar{z}_{v_{f'e'}}.$$

Since  $\psi(f, F, T; \bar{z})$  is maximized by the tree  $T^*$  achieving  $\text{MST}(I_H(F))$ , we obtain

$$\psi_{\max T}(f, F; \bar{z}) := \max_T \psi(f, F, T; \bar{z}) = \psi(f, F, T^*; \bar{z}), \quad (10)$$

that is,

$$\psi(f, F, T^*; \bar{z}) = \sum_{e \in F} \bar{z}_e - S(T^*) - \bar{z}_f + \sum_{v \in P_f} (\bar{z}_v - 1). \quad (11)$$

Therefore, for any fixed pair  $(f, F)$ , determining whether the best spanning-tree inequality is violated reduces to:

Compute  $T^*$  and check whether  $\psi(f, F, T^*; \bar{z}) > 0$ .

This establishes that the inner optimization over  $T$  (hence the test for violation for a fixed pair  $(f, F)$ ) can be performed in polynomial time.

However, the full separation problem requires searching over all subsets  $F \subseteq E \setminus \{f\}$ ; the cardinality of the latter is naively upper bounded by  $|E| * 2^{|E|-1}$ . Actually, it is important to observe that not every subset  $F \subseteq E \setminus \{f\}$  is admissible: in order for a spanning-tree inequality to be well-defined, the partial hypergraph induced by  $F \cup \{f\}$  must contain a  $\gamma$ -cycle. Thus the outer search does not range over all  $2^{|E|-1}$  subsets, but only over those that participate in some  $\gamma$ -cycle with  $f$ . Nevertheless, the number of such cycle-inducing subsets may still be exponential in  $|E|$ , since a hypergraph may contain exponentially many distinct cycles. Consequently, even though the inner optimization over  $T$  is polynomial, the overall separation problem remains combinatorially large.

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